DETAILED DIAGNOSTIC REPORT

COMPREHENSIVE PROPERTY ANALYSIS & ASSESSMENT

■ **AI-Powered Professional Analysis**

■ Generated on:	March 18, 2026 at 10:41 PM
■ Report Type:	Premium DDR Analysis
■ Analysis Method:	Advanced AI Technology

■■ Images Processed:	Inspection: N/A, Thermal: N/A
■ Processing Engine:	Google Gemini 2.5 Flash

Professional Disclaimer: This premium report has been generated using advanced AI analysis of inspection and thermal reports. While our AI system provides comprehensive insights, all findings should be verified by qualified professionals before taking any action. This report represents a sophisticated analysis tool to support expert decision-making.

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**Page numbers are approximate and may vary based on content length*

1. PROPERTY ISSUE SUMMARY

This comprehensive diagnostic report identifies several critical and significant concerns impacting the structural integrity, environmental quality, and energy efficiency of the residential property in Kolkata. The primary findings underscore active moisture intrusion within the bedroom area, manifesting as visible water leakage and mold proliferation, which necessitates immediate attention to mitigate health risks and prevent further material degradation. Concurrently, the living room exhibits signs of compromised building envelope performance, indicated by significant thermal variations and visible cosmetic and structural anomalies. These observations collectively suggest underlying issues requiring prompt and expert intervention to safeguard the property's value and inhabitant wellbeing.

2. AREA-WISE OBSERVATIONS

Bedroom:

Direct inspection of the bedroom ceiling clearly revealed active water leakage, a condition confirmed by the presence of visible mold growth in the affected area. Complementary thermal imaging analysis corroborates this visual finding, presenting distinct cool spots on the ceiling surface, which are highly indicative of moisture saturation. Furthermore, a broader thermal anomaly was detected, strongly suggesting an ongoing seepage issue within the concealed structures of this section of the property. The existence of both active water and mold points to a sustained moisture source requiring urgent investigation.

Living Room:

Within the living room, a visual assessment identified prominent cracks on the

wall surfaces, alongside noticeable paint peeling. These cosmetic deficiencies often signal underlying issues related to wall integrity or moisture ingress. Thermal imaging analysis of this area further highlighted a high degree of temperature variation across the walls, which is a key indicator of potential heat leakage. Such thermal disparities suggest inefficiencies in the building envelope, possibly contributing to energy loss and creating environments that exacerbate material deterioration.

3. PROBABLE ROOT CAUSE

The pervasive moisture and mold issues observed in the bedroom are most likely attributable to an active source of water intrusion. The combination of visible leakage, cool thermal spots, and a pronounced thermal anomaly strongly points towards either a plumbing system failure within overhead or adjacent walls, or a breach in the external building envelope, such as a roof defect or inadequate flashing, allowing rainwater ingress. This sustained moisture creates an ideal environment for microbial proliferation, leading to the detected mold growth.

In the living room, the visible wall cracks and paint peeling, coupled with significant temperature variations and indications of heat leakage, suggest a compromised building envelope. The cracks could stem from structural movement, differential settlement, or issues with plaster adhesion, while the paint peeling often occurs in response to moisture behind the paint film or poor surface preparation. The identified thermal variations and potential heat leakage are symptomatic of insufficient insulation, air infiltration, or thermal bridging within the wall assembly, undermining the thermal performance and contributing to energy inefficiencies.

4. SEVERITY ASSESSMENT

Bedroom Critical:

The issues in the bedroom are deemed Critical due to the immediate and significant risks posed by active water leakage and visible mold growth. Active water intrusion can rapidly escalate structural damage, compromising the integrity of building materials, while mold poses serious health hazards to occupants, particularly those with respiratory sensitivities. The presence of both conditions necessitates prompt, aggressive remediation to prevent further deterioration, extensive repairs, and potential health complications.

Living Room Medium:

The living room concerns are assessed as Medium in severity. While wall cracks and paint peeling are currently primarily aesthetic and indicative of underlying issues, they do suggest potential compromise to the building envelope. The high temperature variation and possible heat leakage contribute to energy inefficiency and increased operational costs, and if unaddressed, could lead to more significant material degradation over time or worsen indoor comfort. These issues require diligent investigation and repair, though they do not present the immediate health and structural risks associated with the bedroom's active water and mold.

5. RECOMMENDED ACTIONS

For the Bedroom, immediate action is paramount. A qualified plumbing and roofing specialist should be engaged without delay to precisely identify and rectify the source of water leakage. Concurrently, a professional mold remediation company must be contracted to safely and thoroughly remove all detected mold growth and to address any compromised building materials. This integrated approach ensures both the cessation of the moisture source and the effective remediation of associated hazards.

For the Living Room, it is recommended to engage a structural engineer or a building envelope specialist to thoroughly assess the nature and cause of the wall cracks and paint peeling. Following this assessment, appropriate repairs for the wall structure and surface finishes should be undertaken. Furthermore, a professional energy audit is advised to precisely quantify the extent of heat leakage and temperature variations, enabling targeted improvements to the building's thermal envelope, such as enhanced insulation or air sealing, to improve energy efficiency and comfort.

6. ADDITIONAL NOTES

The interrelationship between visible defects and thermal anomalies cannot be overstated. The findings within this report underscore the critical importance of utilizing advanced diagnostic tools, such as thermal imaging, to reveal concealed issues that traditional visual inspections alone may not detect. The convergence of visual observations with thermal data provides a much clearer understanding of the root causes, facilitating more effective and durable repair strategies. Proactive and integrated maintenance strategies are essential to preserve property value and ensure a healthy living environment, addressing not only the symptoms but also the underlying causes of building performance deficiencies. Specific building codes or standards relevant to these observations were Not Available within the provided documentation.

7. MISSING OR UNCLEAR INFORMATION

Several critical details were not available within the provided reports, which would further enhance the precision and scope of this diagnostic assessment. The specific historical context of the property, including its age, construction type, and prior renovation history, remains unknown. The exact extent and precise location of the water leakage within the bedroom—for instance, whether it originates from a specific plumbing fixture, a roof penetration, or an external wall crack—are not specified. Similarly, the exact type and approximate area of mold growth were not quantified, which would be beneficial for determining the scope of remediation. For the living room, the specific nature of the wall cracks (e.g., hairline, structural, settlement) and their potential depth or impact on structural elements were not detailed. Furthermore, the magnitude of the "high temperature variation" and "possible heat leakage" identified via thermal imaging lacks specific quantitative metrics, which would allow for a more precise estimation of energy loss. Obtaining this additional information, perhaps through further invasive inspection, moisture mapping, and detailed thermal surveys, would be invaluable for developing the most targeted and costeffective repair strategies.

EXTRACTED IMAGES

Inspection Report Images:

Page 1: inspection_page_1_img_1.jpeg

Thermal Report Images:

Page 1: thermal_page_1_img_1.jpeg

APPENDIX

■ Premium Report Generation Information

This sophisticated Detailed Diagnostic Report has been generated using cutting-edge AI technology combined with advanced building science principles. Our system represents the pinnacle of automated property analysis, delivering insights that traditionally required extensive manual expertise and time investment.

■ Advanced Technical Specifications

• **Analysis Engine:** Google Gemini 2.5 Flash - State-of-the-art AI model
• **PDF Processing:** PyMuPDF Advanced - High-fidelity document extraction
• **Report Generation:** Professional ReportLab Engine - Premium document formatting
• **Processing Date:** 2026-03-18 22:41:19
• **Report Version:** DDR Premium v2.0
• **Quality Assurance:** Multi-layer validation and verification

■ Our Commitment to Excellence

Every report generated through our system undergoes rigorous quality checks and validation processes. We combine the power of artificial intelligence with proven building science methodologies to deliver reports that professionals can trust and clients can understand.

■ Data Processing Capabilities

Our system processes multiple data streams simultaneously, including visual inspection data, thermal imaging information, and environmental factors. This comprehensive approach ensures that no critical information is overlooked and that all findings are properly contextualized within the broader property assessment framework.

■ Continuous Improvement

Our AI models are continuously trained on the latest building science research, industry standards, and real-world inspection data. This ensures that every report reflects the most current understanding of building performance, material

science, and diagnostic methodologies.

Thank you for choosing our premium DDR analysis service. We are committed to providing the highest quality property assessments available in the industry today.